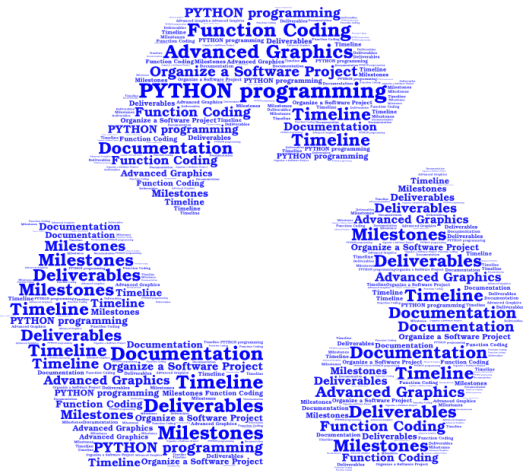




Python Programming: Functions

Lecture: Modelling an Simulation

This is a compulsory elective module
of
**Technology of Circular
Economy**
(Master of Engineering)

Dr. Uwe Reimer
Senior Scientist

Text encryption homework: did you achieve it?

Short summary, what did you learn so far:

- How to run a Python program (overview).
- Code writing (text indent, brackets, comments)

Using functions:

- What is a function?
- The namespace
- Given parameters, global statement, return values

Next week, I am out of office.

Therefore, there will be homework for you.

Challenge from last lesson

The topic was text encryption.

How can you modify your program in order to use any number (not only 3) for the shift in the alphabet?

Did anybody get a solution?

How did you do it?

If not, what was the problem?

Challenge from last lesson

The topic was text encryption.

How can you modify your program in order to use any number (not only 3) for the shift in the alphabet?

You can find the solution (one possible solution) in the Moodle course.

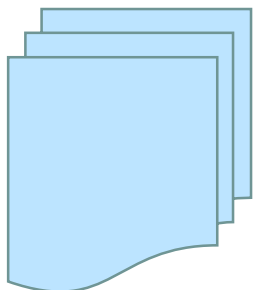
```
10  alorg = 'abcdefghijklmnopqrstuvwxyz'
11  shift = 3    # shift parameter
12  m = ' !'    # non-letter characters in text
13  alnew = alorg[shift:]
14  b = alorg[:shift]
15
16  alnew = alnew + b + m
17  alorg = alorg + m
18  newtext = ''
```



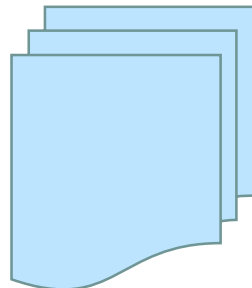
my-program.py

```
1 b = ['beer', 'wine', 'water', 'juice']
2
3 # Say YES to alcohol
4 for c in b:
5     print(c)
6
7     if c == 'beer':
8         print('YES')
9     elif c == 'wine':
10        print('EVEN BETTER')
11    else:
12        print('NO')
13
```

input.txt



output.txt

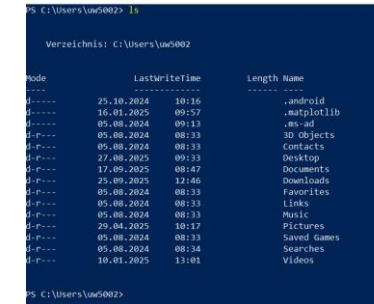


my-run.bat

```
python my-program.py > output.txt
```

Run program by
using either:

*.bat file or
Windows PowerShell





Writing code / syntax

' ' or " " - single or double quotes for strings (both works)

Brackets

() - round bracket or paranthesis → functions

[] - square bracket → position in strings or lists

{ } - curly bracket or brace → dictionaries

Comment

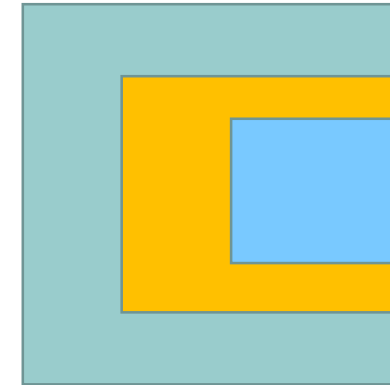
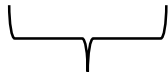
or #: - for comments in a single line

triple quotes ''' or """ - this spans several lines



Block indent

```
1  b = 'hello'
2
3  c = len(b)
4  x = 0
5
6  while x < c:
7      print( b[x] )
8      x = x + 1
9
```



Each new program structure (e.g. loops, if conditions, ...) creates an indent.

The indent needs to be consistent throughout your program. It can be either a tabulator or several blanks.

Normally, your text editor has a default value for this. You may change the default value.

If you copy code from other files, then this must remain consistent.

What is a function?

```
1 b = 'hello'
2
3 c = len(b)
4 x = 0
5
6 while x < c:
7     print( b[x] )
8     x = x + 1
9
```

You can recognize a function by the round brackets ().

A function takes one (or several) arguments and returns a value.

Example:

`c = len(b)`

function = len

argument = b

return value = length of string

What is a function?

```
1 b = 'hello'
2
3 c = len(b)
4 x = 0
5
6 while x < c:
7     print( b[x] )
8     x = x + 1
9
```

You can recognize a function by the round brackets ().

A function takes one (or several) arguments and returns a value.

Example:

`c = len(b)`

function = len

argument = b

return value = length of string

`print( b[x] )`

function = print

argument = b[x] (or a number, a list, ...)

return value = none

In this case the return value is 'none'. The function itself triggers some output to the 'standard output channel'.

-> This works, but it is not useful.

```
1 a = 5
2 b = print(a)
3 print(b)
```

Example:

Create a function that prints a string 3 times.

```
1  b = 'hello'
2
3  def myprint(b):
4      # this function prints a string 3 times
5      print(b)
6      print(b)
7      print(b)
8
9      return None # There is no return value here, yo
10
11     # block indent defines the end of the function
12
13     # here, the function is called (executed)
14     myprint(b)
15
```

This also works. What did change?

```
1  b = 'hello'
2
3  def myprint(x):
4      # this function prints a string 3 times
5      print(x)
6      print(x)
7      print(x)
8
9      return None # There is no return value
10
11     # block indent defines the end of the function
12
13     # here, the function is called (executed)
14     myprint(b)
```

The Namespace

- A namespace is the space where a given name is defined.
- Each function creates a new namespace of it's own. This way it can be avoided that you change a variable (outside the function) by accident.

What will happen here?

```
1  b = 'hello'
2
3  def myprint(x):
4      # this function prints a string 3 times
5      b = 'circular'
6      print(x)
7      print(x)
8      print(x)
9
10     return None # There is no return value here, yo
11
12     # block indent defines the end of the function
13
14     # here, the function is called (executed)
15     myprint(b)
16     print(b)
```

- A namespace is the space where a given name is defined.
- Each function creates a new namespace of it's own. This way it can be avoided that you change a variable (outside the function) by accident.
- You can change that by telling the function to use a variable 'from outside'.
You need to use the 'global' statement.

Check the output of this code.

```
1  b = 'hello'|
2  def myprint(x):
3      # this function prints a string 3 times
4      global b
5
6      b = 'circular'
7      print(x)
8      print(x)
9      print(x)
10     return None # There is no return value
11     # block indent defines the end of the fi
12
13     # here, the function is called (executed)
14     myprint(b)
15     print(b)
```

Example:

Create a function that prints a string x times.

The function needs to take two parameters.

The function should first print the number and then the string, like

1 hello

2 hello

3 hello

...

```
1 b = 'hello'
2 x = 10 # number to repeat string
3
4 def myprint(b,x):
5     # this function prints a string x times
```

Example:

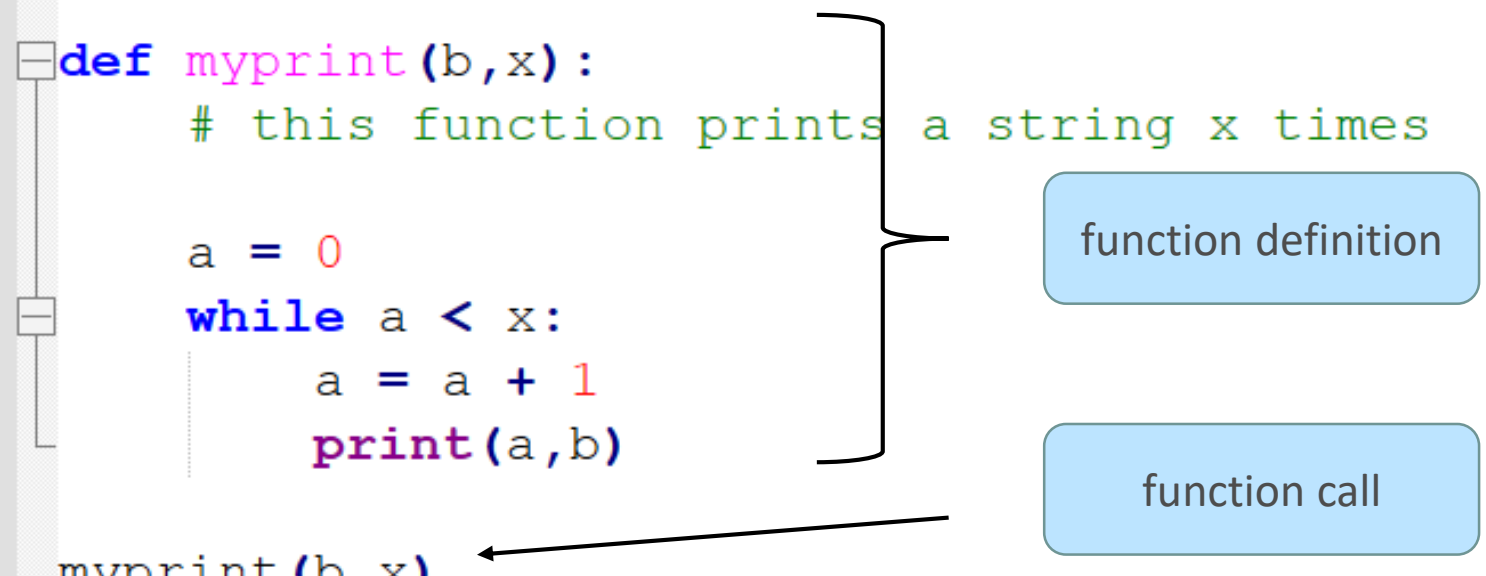
Create a function that prints a string x times.

The function needs to take two parameters.

The function should first print the number and then the string, like

1 hello
2 hello
3 hello
...

```
1  b = 'hello'
2  x = 10 # number to repeat string
3
4  def myprint(b,x):
5      # this function prints a string x times
6
7      a = 0
8      while a < x:
9          a = a + 1
10         print(a,b)
11
12 myprint(b,x)
```



function definition

function call

What did you learn so far from the last lesson:

- Code writing (text indent, brackets
- Run Python programs in *.bat file or from the Windows Powershell
- What is a comment -> which options do you have

What was new:

- What is a function and how do you recognize it?
- What is a namespace and why is it important.
- Given parameters, global statement, return values

From last lesson, we will use 'text encryption – version 3' (see Moodle course).

Your task: Modify the program, i.e. introduce a function that does the text encryption.

The function should take 2 parameters:

- text to encode
- number of shift in alphabet

From last lesson, we will use 'text encryption – version 3' (see Moodle course).

Your task: Modify the program, i.e. introduce a function that does the text encryption.

The function should take 2 parameters:

- text to encode
- number of shift in alphabet

Hint: you could approach the task like this.

- 1) Create a function around the 'for' loops. Be careful with the block indent. This function takes only one parameter: the text to encode. Use the global statement for all other parameters that you need.
- 2) Test your program.
- 3) If this works, add the creation of the new alphabet into the function.
- 4) Test your program.

Solution for the first step:

```
20 def myenc(mytext):
21     # This is a function to encode text.
22     global along, anew, newtext
23
24     for a in mytext:
25         # inside this loop, 'a' contains the next letter from the text
26         # now, find the letter in the original alphabet
27         x = 0
28         for b in along:
29             if b == a: # letter found at position x
30                 newtext = newtext + anew[x] # new text uses same pos
31                 x = x + 1
32
33 myenc(mytext) #This is the function call. If there is no call, the
34
```

Solution for the second step:

```
13 def myenc(mytext, shift):
14     # This is a function to encode text.
15     global newtext
16
17     along = 'abcdefghijklmnopqrstuvwxyz'
18     m = ' !' # non-letter characters in text
19     anew = along[shift:]
20     b = along[:shift]
21
22     anew = anew + b + m
23     along = along + m
24
25     for a in mytext:
26         # inside this loop, 'a' contains the next letter
27         # now, find the letter in the original alphabet
28         x = 0
29         for b in along:
30             if b == a: # letter found at position x
31                 newtext = newtext + anew[x] # new text
32                 x = x + 1
```

You have modified your program to include a function.

- ❑ The function adds some structure to the code.
- ❑ The function should perform a specific task. All necessary data and variables remain encapsulated, i.e. they can not be confused with data outside the function.
- ❑ The 'global' statement gives you an option to exchange data with parts of the code outside the function.
In our specific case, the function could also return the string as return value instead of using the 'global' statement. (Discuss the pro's and con's.)
- ❑ In a way, a function is a small program inside a program.

Optional: Please modify also the decoding program in the same way with a function.

You can also create a function specifically for file input and output.

Task: You will find two encrypted text messages in the moodle course. ☹️

1) (This is for message 1.) Modify your decoding program in a way that it tests for all possible variants of the shift algorithm. Generate text output and decide, which shift was used for encoding.

2) This is for message 2.) Modify your program in order to process text which may also contain capital letters and the characters ' . , ? ' (dot, comma, question mark).

Hint: For capital letters you may check the functions for strings: `string.isupper()` or `string.islower()`

3) Modify your program in the way that it produces an error message, if the original text contains characters that are not recognized. Decide, what your program should do in this case.

It would be cool, if you could at least solve problem number 1). You can work in groups.

